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JEE MAIN · PHYSICS · CHAPTER-WISE PYQS

Mechanical Properties of Solids

JEE Main Physics · Previous Year Questions with Answer Key

130	112	2003 – 2026	English
QUESTIONS	PAPERS COVERED	YEARS	LANGUAGE

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MCQ (Single Correct Answer)	84 questions
Numerical	46 questions
Answer key	at the end

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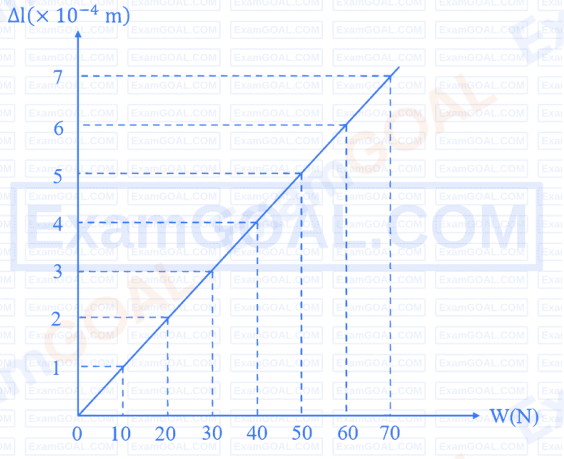
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MCQ (Single Correct Answer) 84 questions

Q1. JEE Main 2026 (Online) 6th April Evening Shift

Open solution

Figure represents the extension (Δl) of a wire of length 1 meter, suspended from the ceiling of the room at one end with a load W connected to the other end. If the cross-sectional area of the wire is $10^{-5}\ \mathrm{m}^2$ then the Young's modulus of the wire is ____ $\mathrm{N/m^2}$.



(A) 1.0×10^{11} (B) 2.0×10^{10} (C) 1.0×10^{10} (D) 2.0×10^{11}

Q2. JEE Main 2026 (Online) 6th April Morning Shift

Open solution

A lift of mass 1600 kg is supported by thick iron wire. If the maximum stress which the wire can withstand is $4 \times 10^8\ \mathrm{N/m^2}$ and its radius is 4 mm , then maximum acceleration the lift can take is ____ $\mathrm{m/s^2}$.

(take $\mathrm{g} = 10\ \mathrm{m/s^2}$ and $\pi = 3.14$)

(A) 2.56 (B) 3.89 (C) 4.32 (D) 5.16

Q3. JEE Main 2026 (Online) 6th April Morning Shift

Open solution

The two wires A and B of equal cross-section but of different materials are joined together. The ratio of Young's modulus of wire A and wire B is 20/11. When the joined wire is kept under certain tension the elongations in the wires A and B are equal. If the length of wire A is 2.2 m , then the length of wire B is

____ m.

(A) 1.1 (B) 2.22 (C) 1.21 (D) 4.44

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Q4. JEE Main 2026 (Online) 5th April Morning Shift

The Young's modulus of steel wire of radius r and length L is Y.

If the radius r and length L of the wire are doubled then the value of Y

(A) increases by two times

(B) reduces by half

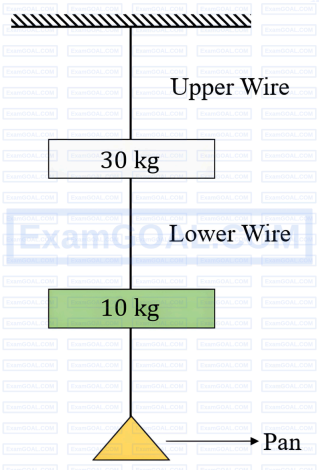
(C) remains unchanged

(D) becomes one fourth

Open solution

Q5. JEE Main 2026 (Online) 5th April Morning Shift

Two wires as shown in the figure below, made of steel and have breaking stress of $12 \times 10^8\ \mathrm{N/m^2}$. Area of cross-section of upper wire is $0.008\ \mathrm{cm}^2$ and of lower wire is $0.004\ \mathrm{cm}^2$. The maximum mass that can be added to pan without breaking any wire is ____ kg.



(A) 56

(B) 38

(C) 96

(D) 5.6

Open solution

Q6. JEE Main 2026 (Online) 4th April Evening Shift

A metal string A is suspended from a rigid support and its free end is attached to a block of mass M. Second block having mass 2 M is suspended at the bottom of the first block using a string B. The area of cross sections of strings A and B are same. The ratio of lengths of strings of A to B is 2 and the ratio of their Young's moduli (Y_A/Y_B) is 0.5 . The ratio of elongations in A to B is ____ .

(A) 1

(B) 4

(C) 8

(D) 6

Open solution

Q7. JEE Main 2026 (Online) 4th April Morning Shift

The increase in the pressure required to decrease the volume (ΔV) of water is $6.3 \times 10^7\ \mathrm{N/m^2}$. The percentage decrease in the volume is ____ .

(Bulk modulus of water $= 2.1 \times 10^9\ \mathrm{N/m^2}$.)

(A) 2 %

(B) 3 %

(C) 6 %

(D) 4 %

Open solution

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Q8. JEE Main 2026 (Online) 4th April Morning Shift

A string A of length 0.314 m and Young's modulus $2 \times 10^{10}\ \mathrm{N/m^2}$ is connected to another string B of length and Young's modulus both twice of those of A. This series combination of strings is then suspended from a rigid support and its free end is fixed to a load of mass 0.8 kg . The net change in length of the combination is ____ mm.

(radius of both the strings is 0.2 mm and acceleration due to gravity $= 10\ \mathrm{m/s^2}$) (Mass of both strings is to be neglected as compared to the mass of load)

Open solution

(A) 3 (B) 2 (C) 1.9 (D) 1

Q9. JEE Main 2026 (Online) 28th January Morning Shift

Two wires A and B made of different materials of lengths 6.0 cm and 5.4 cm , respectively and area of cross sections $3.0 \times 10^{-5}\ \mathrm{m^2}$ and $4.5 \times 10^{-5}\ \mathrm{m^2}$, respectively are stretched by the same magnitude under a given load. The ratio of the Young's modulus of A to that of B is $x : 3$. The value of x is ____

Open solution

(A) 2 (B) 1 (C) 4 (D) 5

Q10. JEE Main 2026 (Online) 24th January Morning Shift

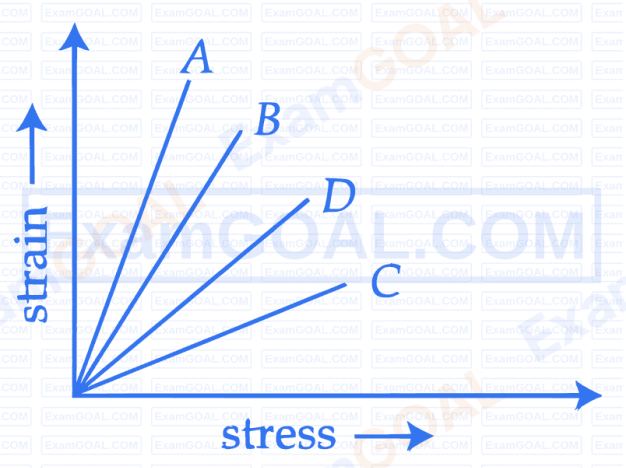
A brass wire of length 2 m and radius 1 mm at $27^\circ\mathrm{C}$ is held taut between two rigid supports. Initially it was cooled to a temperature of $-43^\circ\mathrm{C}$ creating a tension T in the wire. The temperature to which the wire has to be cooled in order to increase the tension in it to $1.4T$, is ____ $^\circ\mathrm{C}$.

Open solution

(A) -71 (B) -80 (C) -65 (D) -86

Q11. JEE Main 2026 (Online) 23rd January Morning Shift

The strain-stress plot for materials A, B, C and D is shown in the figure. Which material has the largest Young's modulus ?



Open solution

(A) B (B) A (C) D (D) C

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Q12. JEE Main 2025 (Online) 8th April Evening Shift

A 3 m long wire of radius 3 mm shows an extension of 0.1 mm when loaded vertically by a mass of 50 kg in an experiment to determine Young's modulus. The value of Young's modulus of the wire as per this experiment is $P \times 10^{11}$ Nm^{-2}, where the value of P is: (Take $g = 3\pi$ m/s^2)

Open solution

(A) 2.5 (B) 25 (C) 10 (D) 5

Q13. JEE Main 2025 (Online) 7th April Morning Shift

Two wires A and B are made of same material having ratio of lengths $\frac{L_A}{L_B} = \frac{1}{3}$ and their diameters ratio $\frac{d_A}{d_B} = 2$. If both the wires are stretched using same force, what would be the ratio of their respective elongations?

Open solution

(A) $3:4$ (B) $1:12$ (C) $1:3$ (D) $1:6$

Q14. JEE Main 2025 (Online) 4th April Evening Shift

A cylindrical rod of length 1 m and radius 4 cm is mounted vertically. It is subjected to a shear force of 10^5 N at the top. Considering infinitesimally small displacement in the upper edge, the angular displacement θ of the rod axis from its original position would be : (shear moduli, $G = 10^{10}$ N/m^2)

Open solution

(A) $1/160\pi$ (B) $1/2\pi$ (C) $1/4\pi$ (D) $1/40\pi$

Q15. JEE Main 2025 (Online) 29th January Morning Shift

The fractional compression $\left(\frac{\Delta V}{V}\right)$ of water at the depth of 2.5 km below the sea level is _____ %. Given, the Bulk modulus of water = 2×10^9 N m^{-2}, density of water = 10^3 kg m^{-3}, acceleration due to gravity $g = 10$ m s^{-2}.

Open solution

(A) 1.0 (B) 1.25 (C) 1.75 (D) 1.5

Q16. JEE Main 2024 (Online) 5th April Evening Shift

Match List I with List II :

Open solution

	LIST I		LIST II
A.	A force that restores an elastic body of unit area to its original state	I.	Bulk modulus
B.	Two equal and opposite forces parallel to opposite faces	II.	Young's modulus
C.	Forces perpendicular everywhere to the surface per unit area same everywhere	III.	Stress
D.	Two equal and opposite forces perpendicular to opposite faces	IV.	Shear modulus

Choose the correct answer from the options given below :

(A) (A)-(IV), (B)-(II), (C)-(III), (D)-(I)
(B) (A)-(III), (B)-(IV), (C)-(I), (D)-(II)
(C) (A)-(II), (B)-(IV), (C)-(I), (D)-(III)
(D) (A)-(III), (B)-(I), (C)-(II), (D)-(IV)

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Q17. JEE Main 2024 (Online) 1st February Morning Shift

With rise in temperature, the Young's modulus of elasticity :

(A) changes erratically
(B) increases
(C) decreases
(D) remains unchanged

Open solution

Q18. JEE Main 2024 (Online) 30th January Morning Shift

Young's modulus of material of a wire of length 'L' and cross-sectional area A is Y. If the length of the wire is doubled and cross-sectional area is halved then Young's modulus will be :

(A) 4 Y
(B) 2 Y
(C) $\frac{Y}{4}$
(D) Y

Open solution

Q19. JEE Main 2024 (Online) 29th January Evening Shift

A wire of length L and radius r is clamped at one end. If its other end is pulled by a force F, its length increases by l. If the radius of the wire and the applied force both are reduced to half of their original values keeping original length constant, the increase in length will become:

(A) 2 times
(B) 4 times
(C) 3 times
(D) $\frac{3}{2}$ times

Open solution

Q20. JEE Main 2024 (Online) 27th January Evening Shift

Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : The property of body, by virtue of which it tends to regain its original shape when the external force is removed, is Elasticity.

Reason (R) : The restoring force depends upon the bonded inter atomic and inter molecular force of solid.

In the light of the above statements, choose the correct answer from the options given below :

(A) (A) is false but (R) is true
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(C) Both (A) and (R) are true and (R) is the correct explanation of (A)
(D) (A) is true but (R) is false

Open solution

Q21. JEE Main 2023 (Online) 15th April Morning Shift

A wire of length ' L ' and radius ' r ' is clamped rigidly at one end. When the other end of the wire is pulled by a force f, its length increases by ' l '. Another wire of same material of length ' $2\,\mathrm{L}$ ' and radius ' $2r$ ' is pulled by a force ' $2f$ '. Then the increase in its length will be :

(A) $2l$
(B) $4l$
(C) l
(D) $l/2$

Open solution

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Q22. JEE Main 2023 (Online) 13th April Morning Shift

Under isothermal condition, the pressure of a gas is given by $\mathrm{P} = a\,\mathrm{V}^{-3}$, where a is a constant and V is the volume of the gas. The bulk modulus at constant temperature is equal to

(A) $\frac{P}{2}$ (B) 2 P (C) 3 P (D) P

Open solution

Q23. JEE Main 2023 (Online) 10th April Evening Shift

Young's moduli of the material of wires A and B are in the ratio of $1:4$, while its area of cross sections are in the ratio of $1:3$. If the same amount of load is applied to both the wires, the amount of elongation produced in the wires A and B will be in the ratio of

[Assume length of wires A and B are same]

(A) 1 : 12 (B) 1 : 36 (C) 12 : 1 (D) 36 : 1

Open solution

Q24. JEE Main 2023 (Online) 8th April Morning Shift

An aluminium rod with Young's modulus $Y = 7.0 \times 10^{10}\ \mathrm{N/m^2}$ undergoes elastic strain of 0.04%. The energy per unit volume stored in the rod in SI unit is:

(A) 5600 (B) 2800 (C) 11200 (D) 8400

Open solution

Q25. JEE Main 2023 (Online) 1st February Evening Shift

The Young's modulus of a steel wire of length $6\ \mathrm{m}$ and cross-sectional area $3\ \mathrm{mm^2}$, is $2 \times 10^{11}\ \mathrm{N/m^2}$. The wire is suspended from its support on a given planet. A block of mass $4\ \mathrm{kg}$ is attached to the free end of the wire. The acceleration due to gravity on the planet is $\frac{1}{4}$ of its value on the earth. The elongation of wire is (Take g on the earth $= 10\ \mathrm{m/s^2}$) :

(A) 0.1 cm (B) 1 cm (C) 0.1 mm (D) 1 mm

Open solution

Q26. JEE Main 2023 (Online) 31st January Evening Shift

Under the same load, wire A having length $5.0\ \mathrm{m}$ and cross section $2.5 \times 10^{-5}\ \mathrm{m^2}$ stretches uniformly by the same amount as another wire B of length $6.0\ \mathrm{m}$ and a cross section of $3.0 \times 10^{-5}\ \mathrm{m^2}$ stretches. The ratio of the Young's modulus of wire A to that of wire B will be :

(A) $1:2$ (B) $1:4$ (C) $1:1$ (D) $1:10$

Open solution

Q27. JEE Main 2023 (Online) 30th January Evening Shift

A force is applied to a steel wire 'A', rigidly clamped at one end. As a result elongation in the wire is $0.2\ \mathrm{mm}$. If same force is applied to another steel wire ' B ' of double the length and a diameter 2.4 times that of the wire ' A ', the elongation in the wire ' B ' will be (wires having uniform circular cross sections)

(A) $6.9 \times 10^{-2}\ \mathrm{mm}$ (B) $6.06 \times 10^{-2}\ \mathrm{mm}$ (C) $2.77 \times 10^{-2}\ \mathrm{mm}$ (D) $3.0 \times 10^{-2}\ \mathrm{mm}$

Open solution

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Q28. JEE Main 2023 (Online) 30th January Morning Shift

Open solution

Choose the correct relationship between Poisson ratio (σ), bulk modulus (K) and modulus of rigidity (η) of a given solid object :

(A) $\sigma = \frac{3K+2\eta}{6K+2\eta}$

(B) $\sigma = \frac{3K-2\eta}{6K+2\eta}$

(C) $\sigma = \frac{6K+2\eta}{3K-2\eta}$

(D) $\sigma = \frac{6K-2\eta}{3K-2\eta}$

Q29. JEE Main 2023 (Online) 24th January Evening Shift

Open solution

Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R

Assertion A : Steel is used in the construction of buildings and bridges.

Reason R : Steel is more elastic and its elastic limit is high.

In the light of above statements, choose the most appropriate answer from the options given below

(A) Both A and R are correct and R is the correct explanation of A

(B) A is correct but R is not correct

(C) Both A and R are correct but R is NOT the correct explanation of A

(D) A is not correct but R is correct

Q30. JEE Main 2023 (Online) 24th January Morning Shift

Open solution

A 100 m long wire having cross-sectional area $6.25 \times 10^{-4}\ \mathrm{m}^2$ and Young's modulus is $10^{10}\ \mathrm{Nm}^{-2}$ is subjected to a load of 250 N, then the elongation in the wire will be :

(A) $6.25 \times 10^{-6}\ \mathrm{m}$

(B) $4 \times 10^{-3}\ \mathrm{m}$

(C) $4 \times 10^{-4}\ \mathrm{m}$

(D) $6.25 \times 10^{-3}\ \mathrm{m}$

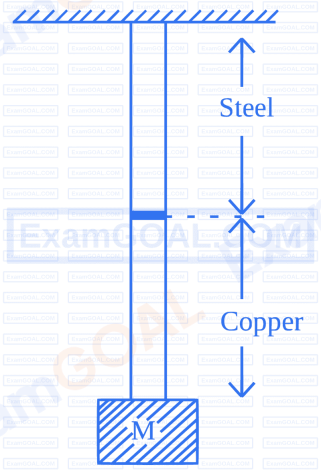
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Q31. JEE Main 2022 (Online) 30th July Evening Shift

Open solution

A block of mass M is attached to a wire. The upper part of wire is made of steel and lower part of wire of copper (as shown in figure). Both parts have the same area of cross section. Neglecting the mass of wires, the ratio of longitudinal strain developed in steel and copper wires is :

(Given, Young's modulus of steel $= 2 \times 10^{11}\ \mathrm{N/m^2}$ and Young's modulus of copper $= 1.5 \times 10^{11}\ \mathrm{N/m^2}$)



(A) $\frac{3}{5}$ (B) $\frac{4}{5}$ (C) $\frac{4}{3}$ (D) $\frac{3}{4}$

Q32. JEE Main 2022 (Online) 29th July Morning Shift

Open solution

If the length of a wire is made double and radius is halved of its respective values. Then, the Young's modulus of the material of the wire will :

(A) remain same
(B) become 8 times its initial value
(C) become $\frac{1}{4}$ of its initial value
(D) become 4 times its initial value

Q33. JEE Main 2022 (Online) 28th July Morning Shift

Open solution

The force required to stretch a wire of cross-section $1\ \mathrm{cm^2}$ to double its length will be : (Given Young's modulus of the wire $= 2 \times 10^{11}\ \mathrm{N/m^2}$)

(A) $1 \times 10^7\ \mathrm{N}$ (B) $1.5 \times 10^7\ \mathrm{N}$ (C) $2 \times 10^7\ \mathrm{N}$ (D) $2.5 \times 10^7\ \mathrm{N}$

Q34. JEE Main 2022 (Online) 27th July Evening Shift

Open solution

A steel wire of length $3.2\ \mathrm{m}$ $\left(\mathrm{Y_s} = 2.0 \times 10^{11}\ \mathrm{Nm^{-2}}\right)$ and a copper wire of length $4.4\ \mathrm{m}$ $\left(\mathrm{Y_c} = 1.1 \times 10^{11}\ \mathrm{Nm^{-2}}\right)$, both of radius $1.4\ \mathrm{mm}$ are connected end to end. When stretched by a load, the net elongation is found to be $1.4\ \mathrm{mm}$. The load applied, in Newton, will be: (Given $\pi = \frac{22}{7}$)

(A) 360 (B) 180 (C) 1080 (D) 154

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Q35. JEE Main 2022 (Online) 26th July Evening Shift

The area of cross section of the rope used to lift a load by a crane is $2.5 \times 10^{-4}\,\mathrm{m}^2$. The maximum lifting capacity of the crane is 10 metric tons. To increase the lifting capacity of the crane to 25 metric tons, the required area of cross section of the rope should be :

(take $g = 10\,ms^{-2}$)

(A) $6.25 \times 10^{-4}\,\mathrm{m}^2$ (B) $10 \times 10^{-4}\,\mathrm{m}^2$

(C) $1 \times 10^{-4}\,\mathrm{m}^2$ (D) $1.67 \times 10^{-4}\,\mathrm{m}^2$

Open solution

Q36. JEE Main 2022 (Online) 29th June Morning Shift

A wire of length L is hanging from a fixed support. The length changes to L_1 and L_2 when masses 1 kg and 2 kg are suspended respectively from its free end. Then the value of L is equal to :

(A) $\sqrt{L_1 L_2}$ (B) $\frac{L_1+L_2}{2}$ (C) $2L_1 - L_2$ (D) $3L_1 - 2L_2$

Open solution

Q37. JEE Main 2022 (Online) 24th June Morning Shift

The bulk modulus of a liquid is 3×10^{10} Nm^{-2}. The pressure required to reduce the volume of liquid by 2% is :

(A) 3×10^{8} Nm^{-2} (B) 9×10^{8} Nm^{-2} (C) 6×10^{8} Nm^{-2} (D) 12×10^{8} Nm^{-2}

Open solution

Q38. JEE Main 2021 (Online) 31st August Evening Shift

Four identical hollow cylindrical columns of mild steel support a big structure of mass 50×10^3 kg. The inner and outer radii of each column are 50 cm and 100 cm respectively. Assuming uniform local distribution, calculate the compression strain of each column. [Use Y = 2.0×10^{11} Pa, g = 9.8 m/s^2]

(A) 3.60×10^{-8} (B) 2.60×10^{-7} (C) 1.87×10^{-3} (D) 7.07×10^{-4}

Open solution

Q39. JEE Main 2021 (Online) 31st August Morning Shift

A uniform heavy rod of weight 10 kg ms^{-2}, cross-sectional area 100 cm^2 and length 20 cm is hanging from a fixed support. Young modulus of the material of the rod is 2×10^{11} Nm^{-2}. Neglecting the lateral contraction, find the elongation of rod due to its own weight.

(A) 2×10^{-9} m (B) 5×10^{-8} m (C) 4×10^{-8} m (D) 5×10^{-10} m

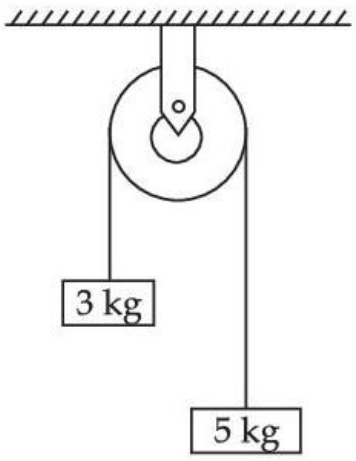
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Q40. JEE Main 2021 (Online) 26th August Evening Shift

Two blocks of masses 3 kg and 5 kg are connected by a metal wire going over a smooth pulley. The breaking stress of the metal is $\frac{24}{\pi} \times 10^2$ Nm^{-2}. What is the minimum radius of the wire ? (Take g = 10 ms^{-2})

Open solution



(A) 125 cm (B) 1250 cm (C) 12.5 cm (D) 1.25 cm

Q41. JEE Main 2021 (Online) 25th July Morning Shift

Two wires of same length and radius are joined end to end and loaded. The Young's moduli of the materials of the two wires are Y_1 and Y_2. The combination behaves as a single wire then its Young's modulus is :

Open solution

(A) $Y = \frac{2Y_1Y_2}{3(Y_1+Y_2)}$ (B) $Y = \frac{2Y_1Y_2}{Y_1+Y_2}$

(C) $Y = \frac{Y_1Y_2}{2(Y_1+Y_2)}$ (D) $Y = \frac{Y_1Y_2}{Y_1+Y_2}$

Q42. JEE Main 2021 (Online) 20th July Evening Shift

The length of a metal wire is l_1, when the tension in it is T_1 and is l_2 when the tension is T_2. The natural length of the wire is :

Open solution

(A) $\sqrt{l_1 l_2}$ (B) $\frac{l_1T_2-l_2T_1}{T_2-T_1}$

(C) $\frac{l_1T_2+l_2T_1}{T_2+T_1}$ (D) $\frac{l_1+l_2}{2}$

Q43. JEE Main 2021 (Online) 20th July Morning Shift

The value of tension in a long thin metal wire has been changed from T_1 to T_2. The lengths of the metal wire at two different values of tension T_1 and T_2 are l_1 and l_2 respectively. The actual length of the metal wire is :

Open solution

(A) $\frac{l_1+l_2}{2}$ (B) $\sqrt{T_1T_2l_1l_2}$

(C) $\frac{T_1l_2-T_2l_1}{T_1-T_2}$ (D) $\frac{T_1l_1-T_2l_2}{T_1-T_2}$

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Q44. JEE Main 2021 (Online) 17th March Evening Shift

Open solution

An object is located at 2 km beneath the surface of the water. If the fractional compression $\frac{\Delta V}{V}$ is 1.36%, the ratio of hydraulic stress to the corresponding hydraulic strain will be _____. [Given : density of water is 1000 kgm^{-3} and g = 9.8 ms^{-2}]

(A) 1.44×10^7 Nm^{-2}

(B) 1.44×10^9 Nm^{-2}

(C) 1.96×10^7 Nm^{-2}

(D) 2.26×10^9 Nm^{-2}

Q45. JEE Main 2021 (Online) 26th February Evening Shift

Open solution

The length of metallic wire is l_1 when tension in it is T_1. It is l_2 when the tension is T_2. The original length of the wire will be :

(A) $\frac{T_1 l_1 - T_2 l_2}{T_2 - T_1}$

(B) $\frac{l_1 + l_2}{2}$

(C) $\frac{T_2 l_1 + T_1 l_2}{T_1 + T_2}$

(D) $\frac{T_2 l_1 - T_1 l_2}{T_2 - T_1}$

Q46. JEE Main 2021 (Online) 26th February Morning Shift

Open solution

The normal density of a material is ρ and its bulk modulus of elasticity is K. The magnitude of increase in density of material, when a pressure P is applied uniformly on all sides, will be :

(A) $\frac{\rho K}{P}$

(B) $\frac{PK}{\rho}$

(C) $\frac{\rho P}{K}$

(D) $\frac{K}{\rho P}$

Q47. JEE Main 2021 (Online) 25th February Morning Shift

Open solution

Given below are two statements : one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A : When a rod lying freely is heated, no thermal stress is developed in it.

Reason R : On heating, the length of the rod increases.

In the light of the above statements, choose the correct answer from the options given below :

(A) A is true but R is false

(B) A is false but R is true

(C) Both A and B are true but R is NOT the correct explanation of A

(D) Both A and R are true and R is the correct explanation of A

Q48. JEE Main 2021 (Online) 24th February Morning Shift

Open solution

If Y, K and η are the values of Young's modulus, bulk modulus and modulus of rigidity of any material respectively. Choose the correct relation for these parameters.

(A) $Y = \frac{9K\eta}{3K-\eta} N/m^2$

(B) $Y = \frac{9K\eta}{2\eta+3K} N/m^2$

(C) $\eta = \frac{3YK}{9K+Y} N/m^2$

(D) $K = \frac{Y\eta}{9\eta-3Y} N/m^2$

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Q49. JEE Main 2020 (Online) 6th September Morning Slot

An object of mass m is suspended at the end of a massless wire of length L and area of crosssection A. Young modulus of the material of the wire is Y. If the mass is pulled down slightly its frequency of oscillation along the vertical direction is :

Open solution

(A) $f = \frac{1}{2\pi}\sqrt{\frac{YA}{mL}}$ (B) $f = \frac{1}{2\pi}\sqrt{\frac{mL}{YA}}$

(C) $f = \frac{1}{2\pi}\sqrt{\frac{YL}{mA}}$ (D) $f = \frac{1}{2\pi}\sqrt{\frac{mA}{YL}}$

Q50. JEE Main 2020 (Online) 4th September Evening Slot

A cube of metal is subjected to a hydrostatic pressure of 4 GPa. The percentage change in the length of the side of the cube is close to :

(Given bulk modulus of metal, B = 8×10^{10} Pa)

Open solution

(A) 0.6 (B) 20 (C) 1.67 (D) 5

Q51. JEE Main 2020 (Online) 9th January Evening Slot

Two steel wires having same length are suspended from a ceiling under the same load. If the ratio of their energy stored per unit volume is 1 : 4, the ratio of their diameters is:

Open solution

(A) 1 : 2 (B) 2 : 1 (C) $1 : \sqrt{2}$ (D) $\sqrt{2} : 1$

Q52. JEE Main 2019 (Online) 12th April Evening Slot

A uniform cylindrical rod of length L and radius r, is made from a material whose Young's modulus of Elasticity equals Y. When this rod is heated by temperature T and simultaneously subjected to a net longitudinal compressional force F, its length remains unchanged. The coefficient of volume expansion, of the material of the rod, is (nearly) equal to :

Open solution

(A) $\frac{3F}{(\pi r^2 YT)}$ (B) $\frac{6F}{(\pi r^2 YT)}$

(C) $\frac{F}{(3\pi r^2 YT)}$ (D) $9F\left(\pi r^2 YT\right)$

Q53. JEE Main 2019 (Online) 12th April Morning Slot

At 40° C, a brass wire of 1 mm radius is hung from the ceiling. A small mass, M is hung from the free end of the wire. When the wire is cooled down from 40°C to 20°C it regains its original length of 0.2 m. The value of M is close to : (Coefficient of linear expansion and Young's modulus of brass are 10^{-5} /^{o}C and 10^{11} N/m ^{2} , respectively; g= 10 ms^{-2})

Open solution

(A) 1.5 kg (B) 0.5 kg (C) 9 kg (D) 0.9 kg

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Q54. JEE Main 2019 (Online) 10th April Evening Slot

In an experiment, brass and steel wires of length 1 m each with areas of cross section 1mm^2 are used. The wires are connected in series and one end of the combined wire is connected to a rigid support and other end is subjected to elongation. The stress required to produce a net elongation of 0.2 mm is, [Given, the Young's Modulus for steel and brass are, respectively, 120×10^9 N/m^2 and 60×10^9 N/m^2]

Open solution

(A) 8.0×10^6 N/m^2 (B) 1.2×10^6 N/m^2 (C) 0.2×10^6 N/m^2 (D) 1.8×10^6 N/m^2

Q55. JEE Main 2019 (Online) 10th April Evening Slot

The elastic limit of brass is 379 MPa. What should be the minimum diameter of a brass rod if it is to support a 400 N load without exceeding its elastic limit?

Open solution

(A) 1.16 mm (B) 1.36 mm (C) 1.00 mm (D) 0.90 mm

Q56. JEE Main 2019 (Online) 8th April Evening Slot

Young's moduli of two wires A and B are in the ratio 7 : 4. Wire A is 2 m long and has radius R. Wire B is 1.5 m long and has radius 2 mm. If the two wires stretch by the same length for a given load, then the value of R is close to :-

Open solution

(A) 1.7 mm (B) 1.9 mm (C) 1.3 mm (D) 1.5 mm

Q57. JEE Main 2019 (Online) 8th April Morning Slot

A boy's catapult is made of rubber cord which is 42 cm long, with 6 mm diameter of cross-section and of negligible mass. The boy keeps a stone weighing 0.02kg on it and stretches the cord by 20 cm by applying a constant force. When released, the stone flies off with a velocity of 20 ms^{-1}. Neglect the change in the area of cross-section of the cord while stretched. The Young's modulus of rubber is closest to:

Open solution

(A) 10^4 Nm^{-2} (B) 10^6 Nm^{-2} (C) 10^8 Nm^{-2} (D) 10^3 Nm^{-2}

Q58. JEE Main 2019 (Online) 8th April Morning Slot

A steel wire having a radius of 2.0 mm, carrying a load of 4 kg, is hanging from a ceiling. Given that g = 3.1π ms^{-2}, what will be the tensile stress that would be developed in the wire ?

Open solution

(A) 3.1×10^6 Nm^{-2} (B) 6.2×10^6 Nm^{-2} (C) 4.8×10^6 Nm^{-2} (D) 5.2×10^6 Nm^{-2}

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Q59. JEE Main 2019 (Online) 12th January Evening Slot

A load of mass M kg is suspended from a steel wire of length 2m and radius 1.0 mm in Searle's apparatus experiment. The increase in length produced in the wire is 4.0 mm. Now the load is fully immersed in a liquid of relative density 2. The relative density of the material of load is 8.

Open solution

The new value of increase in length of the steel wire is:

(A) 5.0 mm (B) zero (C) 3.0 mm (D) 4.0 mm

Q60. JEE Main 2019 (Online) 9th January Morning Slot

A rod, of length L at room temperature and uniform area of cross section A, is made of a metal having coefficient of linear expansion $\alpha/^{o}C$. It is observed that an external compressive force F, is applied on each of its ends, prevents any change in the length of the rod, when its temperature rises by ΔT K. Young's modulus, Y, for this metal is :

Open solution

(A) $\frac{F}{A\alpha\Delta T}$ (B) $\frac{F}{A\alpha(\Delta T-273)}$

(C) $\frac{F}{2A\alpha\Delta T}$ (D) $\frac{2F}{A\alpha\Delta T}$

Q61. JEE Main 2018 (Offline)

A solid sphere of radius r made of a soft material of bulk modulus K is surrounded by a liquid in a cylindrical container. A massless piston of area a floats on the surface of the liquid, covering entire cross section of cylindrical container. When a mass m is placed on the surface of the piston to compress the liquid, the fractional decrement in the radius of the sphere, $\left(\frac{dr}{r}\right)$ is:

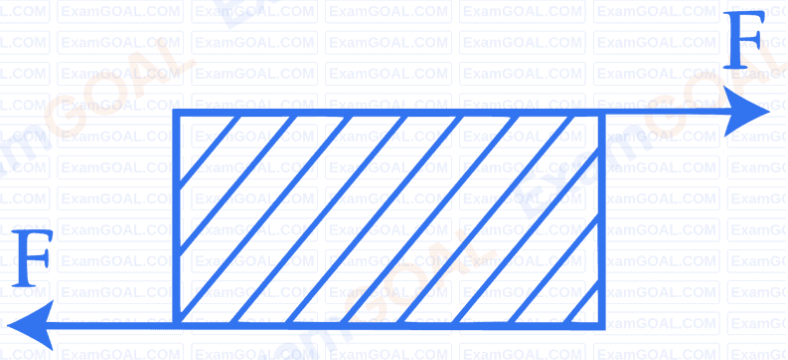
Open solution

(A) $\frac{mg}{Ka}$ (B) $\frac{Ka}{mg}$ (C) $\frac{Ka}{3mg}$ (D) $\frac{mg}{3Ka}$

Q62. JEE Main 2018 (Online) 15th April Evening Slot

As shown in the figure, forces of 10^5 N each are applied in opposite directions, on the upper and lower faces of a cube of side 10 cm, shifting the upper face parallel to itself by 0.5 cm. If the side of another cube of the same material is 20 cm, then under similar conditions as above, the displacement will be :

Open solution



(A) 0.25 cm (B) 0.37 cm (C) 0.75 cm (D) 1.00 cm

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Q63. JEE Main 2017 (Online) 9th April Morning Slot

A steel rail of length 5 m and area of cross section 40cm^2 is prevented from expanding along its length while the temperature rises by 10°C. If coefficient of linear expansion and Young's modulus of steel are 1.2×10^{-5} K^{-1} and 2×10^{11} Nm^{-2} respectively, the force developed in the rail is approximately :

Open solution

(A) 2×10^7 N (B) 1×10^5 N (C) 2×10^9 N (D) 3×10^{-5} N

Q64. JEE Main 2017 (Online) 8th April Morning Slot

A compressive force, F is applied at the two ends of a long thin steel rod. It is heated, simultaneously, such that its temperature increases by ΔT. The net change in its length is zero. Let ℓ be the length of the rod, A its area of cross-section,Y its Young's modulus, and α its coefficient of linear expansion. Then, F is equal to :

Open solution

(A) ℓ^2 Yα ΔT (B) ℓA Yα ΔT (C) A Yα ΔT (D) $\frac{AY}{\alpha \Delta T}$

Q65. JEE Main 2017 (Offline)

A man grows into a giant such that his linear dimensions increase by a factor of 9. Assuming that his density remains same, the stress in the leg will change by a factor of :

Open solution

(A) $\frac{1}{81}$ (B) 9 (C) $\frac{1}{9}$ (D) 81

Q66. JEE Main 2017 (Offline)

An external pressure P is applied on a cube at 0°C so that it is equally compressed from all sides. K is the bulk modulus of the material of the cube and α is its coefficient of linear expansion. Suppose we want to bring the cube to its original size by heating. The temperature should be raised by:

Open solution

(A) $\frac{P}{3\alpha K}$ (B) $\frac{P}{\alpha K}$ (C) $\frac{3\alpha}{PK}$ (D) 3PKα

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Q67. JEE Main 2016 (Online) 10th April Morning Slot

Open solution

A bottle has an opening of radius a and length b. A cork of length b and radius (a + Δa) where (Δa < < a) is compressed to fit into the opening completely (See figure). If the bulk modulus of cork is *B* and frictional coefficient between the bottle and cork is μ then the force needed to push the cork into the bottle is :



(A) ($\pi\,\mu$ B b) Δa

(B) ($2\pi\,\mu$ B b) Δa

(C) ($\pi\,\mu$ B b) a

(D) ($4\pi\,\mu$ B b) Δa

Q68. JEE Main 2016 (Online) 9th April Morning Slot

Open solution

A uniformly tapering conical wire is made from a material of Young's modulus Y and has a normal, unextended length L. The radii, at the upper and lower ends of this conical wire, have values R and 3 R, respectively. The upper end of the wire is fixed to a rigid support and a mass M is suspended from its lower end. The equilibrium extended length, of this wire, would equal :

(A) $L\left(1+\frac{2}{9}\frac{Mg}{\pi YR^2}\right)$

(B) $L\left(1+\frac{1}{3}\frac{Mg}{\pi YR^2}\right)$

(C) $L\left(1+\frac{1}{9}\frac{Mg}{\pi YR^2}\right)$

(D) $L\left(1+\frac{2}{3}\frac{Mg}{\pi YR^2}\right)$

Q69. JEE Main 2014 (Online) 12th April Morning Slot

Open solution

Steel ruptures when a shear of 3.5×10^8 N m^{-2} is applied. The force needed to punch a 1 cm diameter hole in a steel sheet 0.3 cm thick is nearly :

(A) 1.4×10^4 N

(B) 2.7×10^4 N

(C) 3.3×10^4 N

(D) 1.1×10^4 N

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Q70. JEE Main 2014 (Online) 11th April Morning Slot

The Bulk moduli of Ethanol, Mercury and water are given as 0.9, 25 and 2.2 respectively in units of 10^9 Nm^{-2}. For a given value of pressure, the fractional compression in volume is $\frac{\Delta V}{V}$. Which of the following statements about $\frac{\Delta V}{V}$ for these three liquids is correct ?

Open solution

(A) Ethanol > Water > Mercury
(B) Water > Ethanol > Mercury
(C) Mercury > Ethanol > Water
(D) Ethanol > Mercury > Water

Q71. JEE Main 2014 (Online) 9th April Morning Slot

In materials like aluminium and copper, the correct order of magnitude of various elastic moduli is :

Open solution

(A) Young's moduli < shear moduli < bulk moduli.
(B) Bulk moduli < shear moduli < Young's moduli.
(C) Shear moduli < Young's moduli < bulk moduli.
(D) Bulk moduli < Young's moduli < shear moduli

Q72. JEE Main 2014 (Offline)

The pressure that has to be applied to the ends of a steel wire of length $10\ cm$ to keep its length constant when its temperature is raised by $100^{\circ}C$ is: (For steel Young's modulus is $2 \times 10^{11}\ Nm^{-2}$ and coefficient of thermal expansion is $1.1 \times 10^{-5}\ K^{-1}$)

Open solution

(A) $2.2 \times 10^{8}\ Pa$
(B) $2.2 \times 10^{9}\ Pa$
(C) $2.2 \times 10^{7}\ Pa$
(D) $2.2 \times 10^{6}\ Pa$

Q73. JEE Main 2013 (Online) 23th April Morning Slot

A copper wire of length 1.0 m and a steel wire of length 0.5 m having equal cross-sectional areas are joined end to end. The composite wire is stretched by a certain load which stretches the copper wire by 1 mm. If the Young's moduli of copper and steel are respectively 1.0×10^{11} Nm^{-2} and 2.0×10^{11} Nm^{-2}, the total extension of the composite wire is :

Open solution

(A) 1.75 mm
(B) 2.0 mm
(C) 1.50 mm
(D) 1.25 mm

Q74. JEE Main 2013 (Online) 22th April Morning Slot

A uniform wire (Young's modulus 2×10^{11} Nm^{-2}) is subjected to longitudinal tensile stress of 5×10^{7} Nm^{-2}. If the overall volume change in the wire is 0.02%, the fractional decrease in the radius of the wire is close to:

Open solution

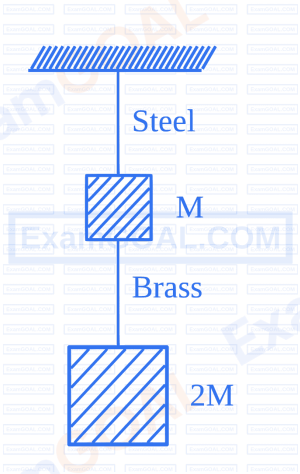
(A) 1.0×10^{-4}
(B) 1.5×10^{-4}
(C) 0.25×10^{-4}
(D) 5×10^{-4}

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Q75. JEE Main 2013 (Online) 9th April Morning Slot

If the ratio of lengths, radii and Young's moduli of steel and brass wires in the figure are a, b and c respectively, then the corresponding ratio of increase in their lengths is :

Open solution



(A) $\frac{3c}{2ab^2}$ (B) $\frac{2a^2c}{b}$ (C) $\frac{3a}{2b^2c}$ (D) $\frac{2ac}{b^2}$

Q76. AIEEE 2012 (Online) 19th May

A steel wire can sustain $100\ \text{kg}$ weight without breaking. If the wire is cut into two equal parts, each part can sustain a weight of

(A) $50\ \text{kg}$ (B) $400\ \text{kg}$ (C) $100\ \text{kg}$ (D) $200\ \text{kg}$

Open solution

Q77. AIEEE 2012 (Online) 7th May

A structural steel rod has a radius of $10\ \text{mm}$ and length of $1.0\ \text{m}$. A $100\ \text{kN}$ force stretches it along its length. Young's modulus of structural steel is $2 \times 10^{11}\ \text{N m}^{-2}$. The percentage strain is about

(A) 0.16% (B) 0.32% (C) 0.08% (D) 0.24%

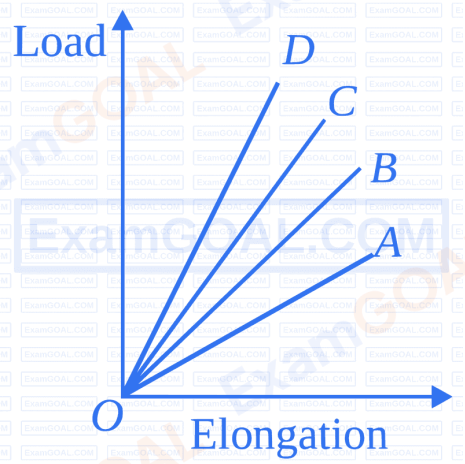
Open solution

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Q78. AIEEE 2012 (Online) 7th May

Open solution

The load versus elongation graphs for four wires of same length and made of the same material are shown in the figure. The thinnest wire is represented by the line

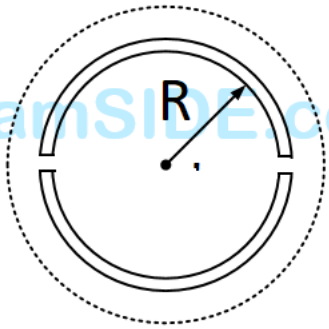


(A) OA (B) OC (C) OD (D) OB

Q79. AIEEE 2012

Open solution

A wooden wheel of radius R is made of two semicircular part (see figure). The two parts are held together by a ring made of a metal strip of cross sectional area S and length L. L is slightly less than $2\pi R$. To fit the ring on the wheel, it is heated so that its temperature rises by ΔT and it just steps over the wheel. As it cools down to surrounding temperature, it process the semicircular parts together. If the coefficient of linear expansion of the metal is α, and its Young's modulus is Y, the force that one part of the wheel applies on the other part is :



(A) $2\pi SY\alpha\Delta T$ (B) $SY\alpha\Delta T$ (C) $\pi SY\alpha\Delta T$ (D) $2SY\alpha\Delta T$

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Q80. AIEEE 2009

Two wires are made of the same material and have the same volume. However wire 1 has cross-sectional area A and wire 2 has cross-sectional area $3A$. If the length of wire 1 increases by Δx on applying force F, how much force is needed to stretch wire 2 by the same amount?

(A) $4F$ (B) $6F$ (C) $9F$ (D) F

Open solution

Q81. AIEEE 2006

A wire elongates by $l\ mm$ when a LOAD W is hanged from it. If the wire goes over a pulley and two weights W each are hung at the two ends, the elongation of the wire will be (in mm)

(A) l (B) $2l$ (C) zero (D) $l/2$

Open solution

Q82. AIEEE 2005

If S is stress and Y is young's modulus of material of a wire, the energy stored in the wire per unit volume is

(A) $\frac{S^2}{2Y}$ (B) $2S^2Y$ (C) $\frac{S}{2Y}$ (D) $\frac{2Y}{S^2}$

Open solution

Q83. AIEEE 2004

A wire fixed at the upper end stretches by length l by applying a force F. The work done in stretching is

(A) $2Fl$ (B) Fl (C) $\frac{F}{2l}$ (D) $\frac{Fl}{2}$

Open solution

Q84. AIEEE 2003

A wire suspended vertically from one of its ends is stretched by attaching a weight of $200N$ to the lower end. The weight stretches the wire by $1\ mm$. Then the elastic energy stored in the wire is

(A) $0.2\ J$ (B) $10\ J$ (C) $20\ J$ (D) $0.1\ J$

Open solution

Numerical 46 questions

Q85. JEE Main 2026 (Online) 6th April Morning Shift

A certain gas is isothermally compressed to $\left(\frac{1}{3}\right)^{\text{rd}}$ of its initial volume ($V_o = 3$ litre) by applying required pressure. If the bulk modulus of the gas is $3 \times 10^5\ \text{N/m}^2$, the magnitude of work done on the gas is ____ J.

Open solution

Q86. JEE Main 2026 (Online) 5th April Evening Shift

A copper wire of length 3 m is stretched by 3 mm by applying an external force. The volume of the wire is $600 \times 10^{-6}\ \text{m}^3$. The elastic potential energy stored in the wire in stretched condition would be ____ J.

(Given Young modulus of copper $= 1.1 \times 10^{11}\ \text{N/m}^2$)

Open solution

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Q87. JEE Main 2026 (Online) 5th April Morning Shift

A cube has side length 5 cm and modulus of rigidity $10^5\ \mathrm{N/m^2}$. The displacement produced by a force of 10 N in the upper face of cube is ____ mm.

Open solution

Q88. JEE Main 2026 (Online) 2nd April Morning Shift

A uniform wire of length l of weight w is suspended from the roof with a weight of W at the other end. The stress in the wire at $\frac{l}{3}$ distance from the top is $\left(\frac{W}{A}+\frac{2}{\gamma}\frac{w}{A}\right)$, where A is the cross sectional area of the wire. The value of γ is _____.

Open solution

Q89. JEE Main 2025 (Online) 8th April Evening Shift

A sample of a liquid is kept at 1 atm. It is compressed to 5 atm which leads to a change of volume of 0.8 cm^3. If the bulk modulus of the liquid is 2 GPa, the initial volume of the liquid was _____ litre.

(Take 1 atm = 10^5 Pa)

Open solution

Q90. JEE Main 2025 (Online) 4th April Morning Shift

Two slabs with square cross section of different materials $(1, 2)$ with equal sides (l) and thickness d_1 and d_2 such that $d_2 = 2d_1$ and $l > d_2$. Considering lower edges of these slabs are fixed to the floor, we apply equal shearing force on the narrow faces. The angle of deformation is $\theta_2 = 2\theta_1$. If the shear moduli of material 1 is $4 \times 10^9\ \mathrm{N/m^2}$, then shear moduli of material 2 is $x \times 10^9\ \mathrm{N/m^2}$, where value of x is _____.

Open solution

Q91. JEE Main 2025 (Online) 2nd April Evening Shift

The length of a light string is 1.4 m when the tension on it is 5 N . If the tension increases to 7 N , the length of the string is 1.56 m . The original length of the string is_____m.

Open solution

Q92. JEE Main 2025 (Online) 2nd April Morning Shift

A steel wire of length 2 m and Young's modulus $2.0 \times 10^{11}\ \mathrm{N\,m^{-2}}$ is stretched by a force. If Poisson ratio and transverse strain for the wire are 0.2 and 10^{-3} respectively, then the elastic potential energy density of the wire is _____ $\times 10^5$ (in SI units).

Open solution

Q93. JEE Main 2025 (Online) 28th January Evening Shift

The volume contraction of a solid copper cube of edge length 10 cm , when subjected to a hydraulic pressure of $7 \times 10^6\ \mathrm{Pa}$, would be _____ $\mathrm{mm^3}$.

(Given bulk modulus of copper $= 1.4 \times 10^{11}\ \mathrm{N\,m^{-2}}$)

Open solution

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Q94. JEE Main 2025 (Online) 24th January Evening Shift

The increase in pressure required to decrease the volume of a water sample by 0.2% is $\mathrm{P} \times 10^5 \mathrm{Nm}^{-2}$. Bulk modulus of water is $2.15 \times 10^9 \mathrm{Nm}^{-2}$. The value of P is _____ .

Open solution

Q95. JEE Main 2024 (Online) 9th April Morning Shift

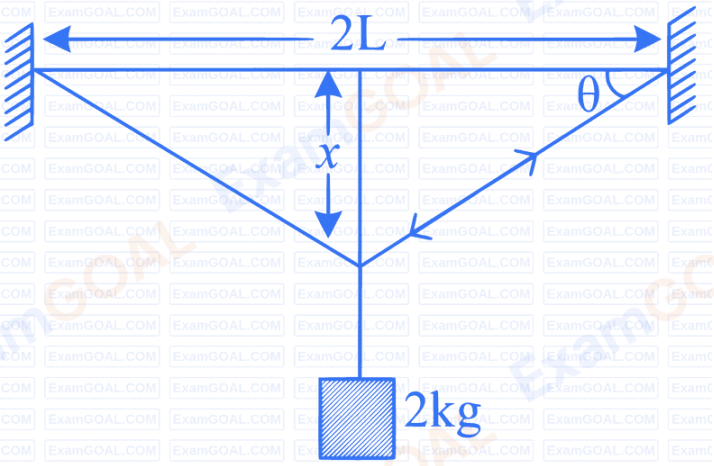
Two persons pull a wire towards themselves. Each person exerts a force of $200\ \mathrm{N}$ on the wire. Young's modulus of the material of wire is $1 \times 10^{11}\ \mathrm{N\,m^{-2}}$. Original length of the wire is $2\ \mathrm{m}$ and the area of cross section is $2\ \mathrm{cm}^2$. The wire will extend in length by _____ $\mu\mathrm{m}$.

Open solution

Q96. JEE Main 2024 (Online) 6th April Evening Shift

A wire of cross sectional area A, modulus of elasticity $2 \times 10^{11}\ \mathrm{Nm}^{-2}$ and length $2\ \mathrm{m}$ is stretched between two vertical rigid supports. When a mass of $2\ \mathrm{kg}$ is suspended at the middle it sags lower from its original position making angle $\theta = \frac{1}{100}$ radian on the points of support. The value of A is _____ $\times 10^{-4}\ \mathrm{m}^2$ (consider $x << \mathrm{L}$).

(given : $\mathrm{g} = 10\ \mathrm{m/s}^2$)



Open solution

Q97. JEE Main 2024 (Online) 5th April Morning Shift

The density and breaking stress of a wire are $6 \times 10^4\ \mathrm{kg/m}^3$ and $1.2 \times 10^8\ \mathrm{N/m}^2$ respectively. The wire is suspended from a rigid support on a planet where acceleration due to gravity is $\frac{1}{3}^{\mathrm{rd}}$ of the value on the surface of earth. The maximum length of the wire without breaking is _____ m (take, $\mathrm{g} = 10\ \mathrm{m/s}^2$).

Open solution

Q98. JEE Main 2024 (Online) 4th April Morning Shift

An elastic spring under tension of $3\ \mathrm{N}$ has a length a. Its length is b under tension $2\ \mathrm{N}$. For its length $(3a - 2b)$, the value of tension will be _____ N.

Open solution

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Q99. JEE Main 2024 (Online) 1st February Evening Shift

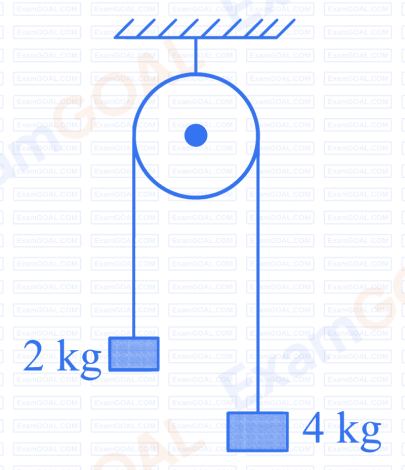
One end of a metal wire is fixed to a ceiling and a load of $2\,\text{kg}$ hangs from the other end. A similar wire is attached to the bottom of the load and another load of $1\,\text{kg}$ hangs from this lower wire. Then the ratio of longitudinal strain of upper wire to that of the lower wire will be _____.

[Area of cross section of wire $= 0.005\,\text{cm}^2$, $\text{Y} = 2 \times 10^{11}\text{Nm}^{-2}$ and $\text{g} = 10\,\text{ms}^{-2}$]

Open solution

Q100. JEE Main 2024 (Online) 31st January Evening Shift

Two blocks of mass $2\,\text{kg}$ and $4\,\text{kg}$ are connected by a metal wire going over a smooth pulley as shown in figure. The radius of wire is $4.0 \times 10^{-5}\,\text{m}$ and Young's modulus of the metal is $2.0 \times 10^{11}\,\text{N/m}^2$. The longitudinal strain developed in the wire is $\frac{1}{\alpha\pi}$. The value of α is _____. [Use $g = 10\,\text{m/s}^2$]



Open solution

Q101. JEE Main 2024 (Online) 31st January Morning Shift

The depth below the surface of sea to which a rubber ball be taken so as to decrease its volume by 0.02% is _____ m.

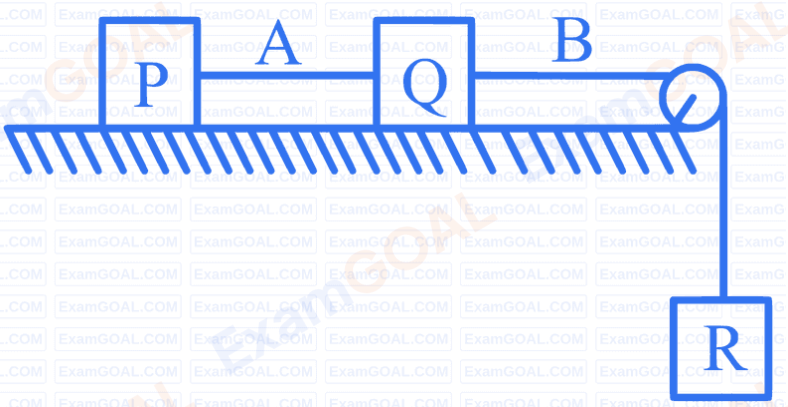
(Take density of sea water $= 10^3\text{kgm}^{-3}$, Bulk modulus of rubber $= 9 \times 10^8\,\text{Nm}^{-2}$, and $g = 10\,\text{ms}^{-2}$)

Open solution

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Q102. JEE Main 2024 (Online) 30th January Morning Shift

Each of three blocks P, Q and R shown in figure has a mass of $3\ \mathrm{kg}$. Each of the wires A and B has cross-sectional area $0.005\ \mathrm{cm}^2$ and Young's modulus $2 \times 10^{11}\ \mathrm{N\,m^{-2}}$. Neglecting friction, the longitudinal strain on wire B is _____ $\times 10^{-4}$. (Take $\mathrm{g} = 10\ \mathrm{m/s^2}$)



Open solution

Q103. JEE Main 2024 (Online) 29th January Evening Shift

Two metallic wires P and Q have same volume and are made up of same material. If their area of cross sections are in the ratio $4 : 1$ and force F_1 is applied to P, an extension of Δl is produced. The force which is required to produce same extension in Q is F_2.

The value of $\frac{F_1}{F_2}$ is _____.

Open solution

Q104. JEE Main 2024 (Online) 27th January Morning Shift

If average depth of an ocean is $4000\ \mathrm{m}$ and the bulk modulus of water is $2 \times 10^9\ \mathrm{Nm^{-2}}$, then fractional compression $\frac{\Delta V}{V}$ of water at the bottom of ocean is $\alpha \times 10^{-2}$. The value of α is _____ (Given, $\mathrm{g} = 10\ \mathrm{ms^{-2}}$, $\rho = 1000\ \mathrm{kg\,m^{-3}}$)

Open solution

Q105. JEE Main 2023 (Online) 13th April Morning Shift

The elastic potential energy stored in a steel wire of length $20\ \mathrm{m}$ stretched through $2\ \mathrm{cm}$ is $80\ \mathrm{J}$. The cross sectional area of the wire is _____ mm^2.

(Given, $Y = 2.0 \times 10^{11}\mathrm{Nm^{-2}}$)

Open solution

Q106. JEE Main 2023 (Online) 11th April Morning Shift

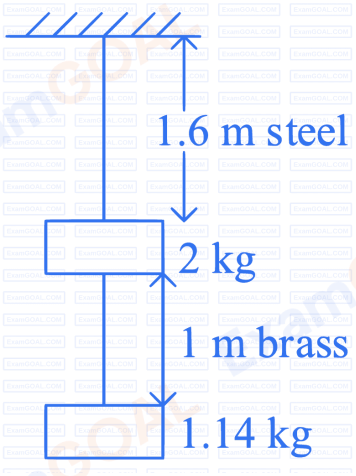
The length of a wire becomes l_1 and l_2 when $100\ \mathrm{N}$ and $120\ \mathrm{N}$ tensions are applied respectively. If $10\,l_2 = 11\,l_1$, the natural length of wire will be $\frac{1}{x}\,l_1$. Here the value of x is _____.

Open solution

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Q107. JEE Main 2023 (Online) 10th April Morning Shift

Two wires each of radius 0.2 cm and negligible mass, one made of steel and the other made of brass are loaded as shown in the figure. The elongation of the steel wire is _____ $\times 10^{-6}$ m. [Young's modulus for steel = 2×10^{11} Nm^{-2} and g = 10 ms^{-2}]



Open solution

Q108. JEE Main 2023 (Online) 8th April Evening Shift

A steel rod of length $1\ \mathrm{m}$ and cross sectional area $10^{-4}\ \mathrm{m}^2$ is heated from $0^{\circ}\mathrm{C}$ to $200^{\circ}\mathrm{C}$ without being allowed to extend or bend. The compressive tension produced in the rod is _____ $\times 10^4\ \mathrm{N}$. (Given Young's modulus of steel $= 2 \times 10^{11}\mathrm{Nm}^{-2}$, coefficient of linear expansion $= 10^{-5}\ \mathrm{K}^{-1}$)

Open solution

Q109. JEE Main 2023 (Online) 6th April Evening Shift

A metal block of mass m is suspended from a rigid support through a metal wire of diameter $14\ \mathrm{mm}$. The tensile stress developed in the wire under equilibrium state is $7 \times 10^5 \mathrm{Nm}^{-2}$. The value of mass m is _____ kg. (Take, $\mathrm{g} = 9.8\ \mathrm{ms}^{-2}$ and $\pi = \frac{22}{7}$)

Open solution

Q110. JEE Main 2023 (Online) 6th April Morning Shift

A steel rod has a radius of $20\ \mathrm{mm}$ and a length of $2.0\ \mathrm{m}$. A force of $62.8\ \mathrm{kN}$ stretches it along its length. Young's modulus of steel is $2.0 \times 10^{11}\ \mathrm{N/m}^2$. The longitudinal strain produced in the wire is _____ $\times 10^{-5}$

Open solution

Q111. JEE Main 2023 (Online) 1st February Morning Shift

A certain pressure 'P' is applied to 1 litre of water and 2 litre of a liquid separately. Water gets compressed to 0.01% whereas the liquid gets compressed to 0.03%. The ratio of Bulk modulus of water to that of the liquid is $\frac{3}{x}$. The value of x is _____.

Open solution

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Q112. JEE Main 2023 (Online) 31st January Morning Shift

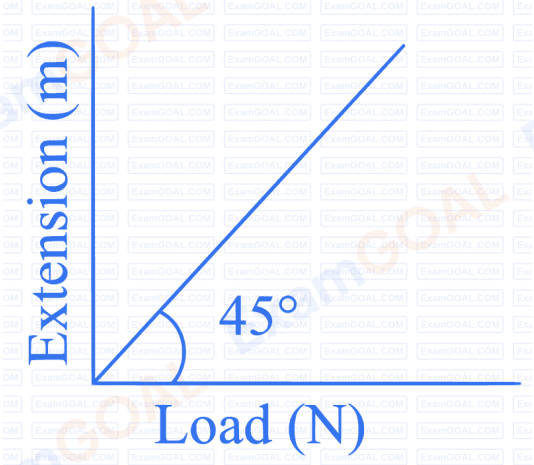
A thin rod having a length of $1\ \mathrm{m}$ and area of cross-section $3 \times 10^{-6}\ \mathrm{m}^2$ is suspended vertically from one end. The rod is cooled from $210^{\circ}\mathrm{C}$ to $160^{\circ}\mathrm{C}$. After cooling, a mass M is attached at the lower end of the rod such that the length of rod again becomes $1\ \mathrm{m}$. Young's modulus and coefficient of linear expansion of the rod are $2 \times 10^{11}\ \mathrm{N\,m^{-2}}$ and $2 \times 10^{-5}\ \mathrm{K^{-1}}$, respectively. The value of M is _____ kg.

(Take $\mathrm{g} = 10\ \mathrm{m\,s^{-2}}$)

Open solution

Q113. JEE Main 2023 (Online) 25th January Morning Shift

As shown in the figure, in an experiment to determine Young's modulus of a wire, the extension-load curve is plotted. The curve is a straight line passing through the origin and makes an angle of 45° with the load axis. The length of wire is 62.8 cm and its diameter is 4 mm. The Young's modulus is found to be $x \times 10^4$ $\mathrm{Nm^{-2}}$. The value of x is _____.



Open solution

Q114. JEE Main 2022 (Online) 29th July Evening Shift

A metal wire of length $0.5\ \mathrm{m}$ and cross-sectional area $10^{-4}\ \mathrm{m}^2$ has breaking stress $5 \times 10^8\ \mathrm{Nm^{-2}}$. A block of $10\ \mathrm{kg}$ is attached at one end of the string and is rotating in a horizontal circle. The maximum linear velocity of block will be _____ $\mathrm{ms^{-1}}$.

Open solution

Q115. JEE Main 2022 (Online) 29th July Evening Shift

The speed of a transverse wave passing through a string of length $50\ \mathrm{cm}$ and mass $10\ \mathrm{g}$ is $60\ \mathrm{ms^{-1}}$. The area of cross-section of the wire is $2.0\ \mathrm{mm}^2$ and its Young's modulus is $1.2 \times 10^{11}\mathrm{Nm^{-2}}$. The extension of the wire over its natural length due to its tension will be $x \times 10^{-5}\ \mathrm{m}$. The value of x is _____.

Open solution

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Q116. JEE Main 2022 (Online) 28th July Evening Shift

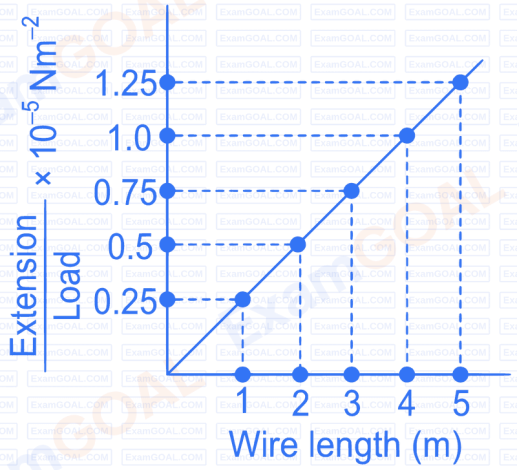
A string of area of cross-section $4\ \mathrm{mm}^2$ and length $0.5\ \mathrm{m}$ is connected with a rigid body of mass $2\ \mathrm{kg}$. The body is rotated in a vertical circular path of radius $0.5\ \mathrm{m}$. The body acquires a speed of $5\ \mathrm{m/s}$ at the bottom of the circular path. Strain produced in the string when the body is at the bottom of the circle is _____ $\times 10^{-5}$.

(use Young's modulus $10^{11}\ \mathrm{N/m^2}$ and $\mathrm{g} = 10\ \mathrm{m/s^2}$)

Open solution

Q117. JEE Main 2022 (Online) 27th July Evening Shift

In an experiment to determine the Young's modulus, steel wires of five different lengths $(1, 2, 3, 4,$ and $5\ \mathrm{m})$ but of same cross section $\left(2\ \mathrm{mm}^2\right)$ were taken and curves between extension and load were obtained. The slope (extension/load) of the curves were plotted with the wire length and the following graph is obtained. If the Young's modulus of given steel wires is $x \times 10^{11}\ \mathrm{Nm}^{-2}$, then the value of x is _____.



Open solution

Q118. JEE Main 2022 (Online) 27th July Morning Shift

A square aluminum (shear modulus is $25 \times 10^9\ \mathrm{Nm}^{-2}$) slab of side $60\ \mathrm{cm}$ and thickness $15\ \mathrm{cm}$ is subjected to a shearing force (on its narrow face) of $18.0 \times 10^4\ \mathrm{N}$. The lower edge is riveted to the floor. The displacement of the upper edge is _____ μm.

Open solution

Q119. JEE Main 2022 (Online) 26th July Evening Shift

A uniform heavy rod of mass $20\ \mathrm{kg}$, cross sectional area $0.4\ \mathrm{m}^2$ and length $20\ \mathrm{m}$ is hanging from a fixed support. Neglecting the lateral contraction, the elongation in the rod due to its own weight is $x \times 10^{-9}\ \mathrm{m}$. The value of x is _____.

(Given, young modulus Y = $2 \times 10^{11}\ \mathrm{Nm}^{-2}$ and g = $10\ \mathrm{ms}^{-2}$)

Open solution

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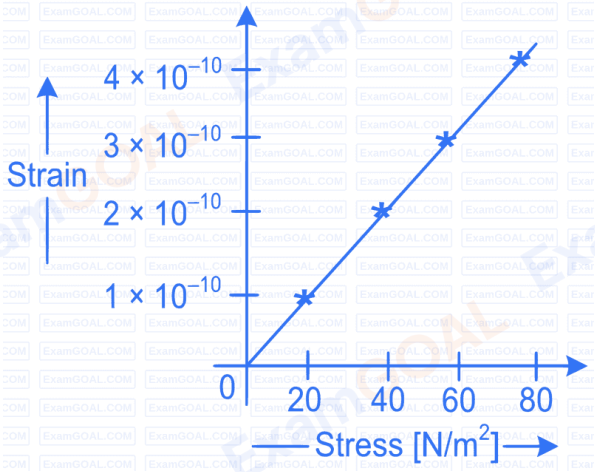
Q120. JEE Main 2022 (Online) 25th July Morning Shift

A wire of length L and radius r is clamped rigidly at one end. When the other end of the wire is pulled by a force F, its length increases by $5\ cm$. Another wire of the same material of length $4L$ and radius $4r$ is pulled by a force $4F$ under same conditions. The increase in length of this wire is _____ cm.

Open solution

Q121. JEE Main 2022 (Online) 26th June Morning Shift

The elastic behaviour of material for linear stress and linear strain, is shown in the figure. The energy density for a linear strain of 5×10^{-4} is _____ kJ/m^3. Assume that material is elastic upto the linear strain of 5×10^{-4}.



Open solution

Q122. JEE Main 2022 (Online) 26th June Morning Shift

The elongation of a wire on the surface of the earth is 10^{-4} m. The same wire of same dimensions is elongated by 6×10^{-5} m on another planet. The acceleration due to gravity on the planet will be _____ ms^{-2}. (Take acceleration due to gravity on the surface of earth = 10 ms^{-2})

Open solution

Q123. JEE Main 2021 (Online) 1st September Evening Shift

A steel rod with y = 2.0×10^{11} Nm^{-2} and $\alpha = 10^{-5}\ ^\circ C^{-1}$ of length 4 m and area of cross-section 10 cm^2 is heated from 0°C to 400°C without being allowed to extend. The tension produced in the rod is $x \times 10^5$ N where the value of x is _____.

Open solution

Q124. JEE Main 2021 (Online) 31st August Morning Shift

When a rubber ball is taken to a depth of _____m in deep sea, its volume decreases by 0.5%.

(The bulk modulus of rubber = 9.8×10^8 Nm^{-2}, Density of sea water = 10^3 kgm^{-3}, g = 9.8 m/s^2)

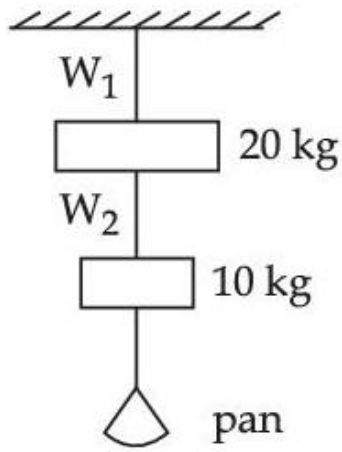
Open solution

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Q125. JEE Main 2021 (Online) 27th August Evening Shift

Open solution

Wires W_1 and W_2 are made of same material having the breaking stress of 1.25×10^9 N/m^2. W_1 and W_2 have cross-sectional area of 8×10^{-7} m^2 and 4×10^{-7} m^2, respectively. Masses of 20 kg and 10 kg hang from them as shown in the figure. The maximum mass that can be placed in the pan without breaking the wires is _____ kg. (Use g = 10 m/s^2)



Q126. JEE Main 2021 (Online) 27th July Morning Shift

Open solution

A stone of mass 20 g is projected from a rubber catapult of length 0.1 m and area of cross section 10^{-6} m^2 stretched by an amount 0.04 m. The velocity of the projected stone is _____ m/s.

(Young's modulus of rubber = 0.5×10^9 N/m^2)

Q127. JEE Main 2021 (Online) 22th July Evening Shift

Open solution

The area of cross-section of a railway track is 0.01 m^2. The temperature variation is 10°C. Coefficient of linear expansion of material of track is 10^{-5}/°C. The energy stored per meter in the track is _____ J/m.

(Young's modulus of material of track is 10^{11} Nm^{-2})

Q128. JEE Main 2021 (Online) 18th March Morning Shift

Open solution

Two separate wires A and B are stretched by 2 mm and 4 mm respectively, when they are subjected to a force of 2 N. Assume that both the wires are made up of same material and the radius of wire B is 4 times that of the radius of wire A. The length of the wires A and B are in the ratio of a : b. Then a/b can be expressed as 1/x where x is _____.

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Q129. JEE Main 2021 (Online) 24th February Evening Shift

A uniform metallic wire is elongated by 0.04 m when subjected to a linear force F. The elongation, if its length and diameter is doubled and subjected to the same force will be _____ cm.

Open solution

Q130. JEE Main 2020 (Online) 9th January Morning Slot

A body of mass m = 10 kg is attached to one end of a wire of length 0.3 m. The maximum angular speed (in rad s^{-1}) with which it can be rotated about its other end in space station is :
(Breaking stress of wire = 4.8×10^7 Nm^{-2} and
area of cross-section of the wire = 10^{-2} cm^2) is:

Open solution

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Answer Key

MCQ (Single Correct Answer)

1. **C**	2. **A**	3. **C**	4. **C**	5. **B**
6. **D**	7. **B**	8. **B**	9. **D**	10. **A**
11. **D**	12. **D**	13. **B**	14. **A**	15. **B**
16. **B**	17. **C**	18. **D**	19. **A**	20. **C**
21. **C**	22. **C**	23. **C**	24. **A**	25. **C**
26. **C**	27. **A**	28. **B**	29. **A**	30. **B**
31. **D**	32. **A**	33. **C**	34. **D**	35. **A**
36. **C**	37. **C**	38. **B**	39. **D**	40. **C**
41. **B**	42. **B**	43. **C**	44. **B**	45. **D**
46. **C**	47. **C**	48. **D**	49. **A**	50. **C**
51. **D**	52. **A**	53. **C**	54. **A**	55. **A**
56. **A**	57. **B**	58. **A**	59. **C**	60. **A**
61. **D**	62. **A**	63. **B**	64. **C**	65. **B**
66. **A**	67. **D**	68. **B**	69. **C**	70. **A**
71. **C**	72. **A**	73. **D**	74. **C**	75. **C**
76. **C**	77. **A**	78. **A**	79. **D**	80. **C**
81. **A**	82. **A**	83. **D**	84. **D**	

Numerical

85. **989**	86. **33**	87. **2**	88. **3**	89. **4**
90. **1**	91. **1**	92. **25**	93. **50**	94. **43**
95. **20**	96. **1**	97. **600**	98. **5**	99. **3**
100. **12**	101. **18**	102. **2**	103. **16**	104. **2**
105. **40**	106. **2**	107. **20**	108. **4**	109. **11**
110. **25**	111. **1**	112. **60**	113. **5**	114. **50**
115. **15**	116. **30**	117. **2**	118. **48**	119. **25**
120. **5**	121. **25**	122. **6**	123. **8**	124. **500**
125. **40**	126. **20**	127. **05**	128. **32**	129. **2**
130. **4**				

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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